Large Deviations of Fréchet Means on Stratified Metric Spaces

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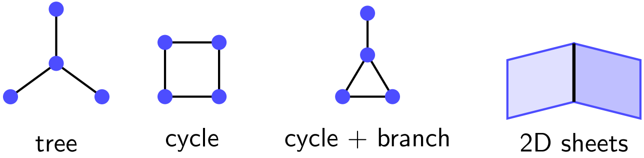
Stratified Spaces: Pieces Glued Together

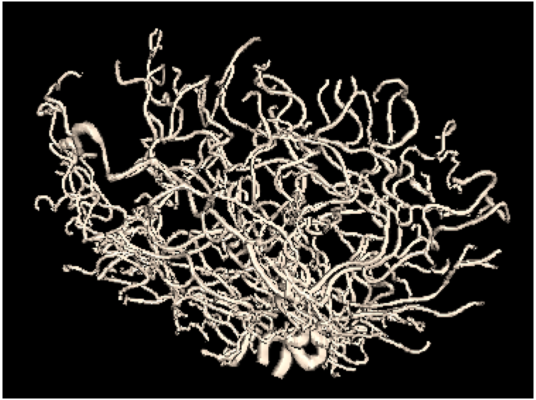
A **stratified space** is built by gluing together simple pieces, such as line segments, surfaces, or higher-dimensional regions.

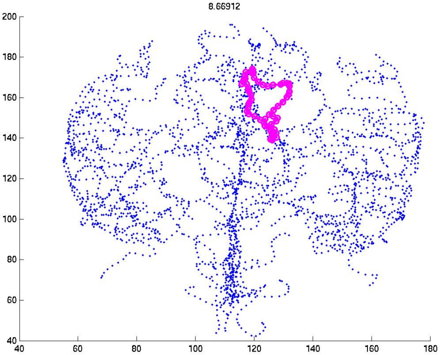


Distance = length of shortest path inside the glued space.

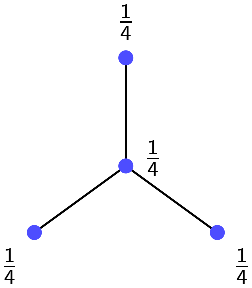
Why This Space? A Real-World Motivation

Many real datasets branch, like arteries or the brain's blood vessels. These naturally live on stratified spaces.



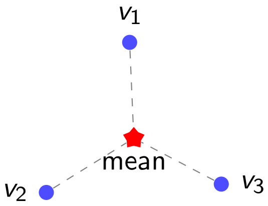


The Uniform Distribution



Draw N points $X_1, \ldots, X_N$, each landing on a vertex independently and uniformly at random.

Fréchet Mean: the "Center" of a Set of Points



In Euclidean space, we average the points directly:

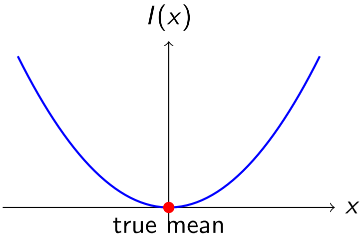
$$\text{mean} = \frac{1}{N}\sum_i v_i.$$

The Fréchet mean generalizes this to the point minimizing total squared distance:

$$\text{mean} = \arg\min_t \sum_i d(t, v_i)^2.$$

It works in *any* metric space — no addition required.

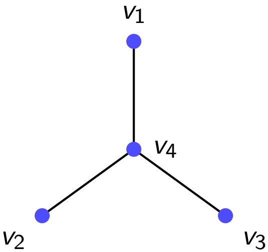
Large Deviations: How Rare is a Deviation of the Mean?



Cramér's theorem: the sample mean strays from the truth with probability

$$\mathbb{P} \approx e^{-N \cdot I(t)}.$$

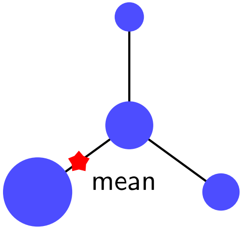
Our Problem



On a stratified metric space, sample N points uniformly on the vertices.

Goal: relate qualitative properties of the rate function $I(t)$ to geometric properties of the space.

Finding the Rate Function



The Fréchet mean is determined entirely by the **proportion of points** at each vertex, $\mathbf{p} = (p_1, \ldots, p_n)$.

Step 1: rate function for how far $\mathbf{p}$ strays from uniform (Sanov's Theorem).

Step 2: contraction principle carries this over to the Fréchet mean — rare values of $\mathbf{p}$ map to rare values of the mean $t = \mu(\mathbf{p})$, at the same exponential rate.

So the Problem Boils Down To...

Minimize divergence from the uniform distribution, subject to landing on t:

$$I(t) = \inf_{\mathbf{p}\,:\,\mu(\mathbf{p})=t} D_{\mathrm{KL}}(\mathbf{p} \,\|\, \mathbf{u}), \qquad \text{where } D_{\mathrm{KL}}(\mathbf{p} \,\|\, \mathbf{u}) = \sum_k p_k \log(N p_k).$$

A constrained optimization problem.

Theorem: 1D Stratified Spaces

Let G be a connected 1D stratified metric space with vertex set V, and let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} \mathrm{Unif}(V)$. Let $I(t)$ be the large deviation rate function of the Fréchet mean of $X_1, \ldots, X_N$.

Theorem (Informal)

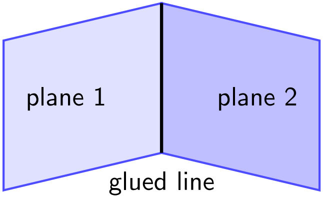
$I(t)$ is continuous on G, and smooth except at exactly two kinds of points:

- **vertices** *of degree ≥ 3 — a* **jump***;*
- **switching points** *where the shortest route to some vertex flips direction (only when loops are present) — a* **cusp***.*

Corollary: stickiness

If a jump vertex is also the population Fréchet mean, i.e., $I(v) = 0$, then the empirical Fréchet mean is **sticky** there: moving into any incident edge has positive large-deviation cost.

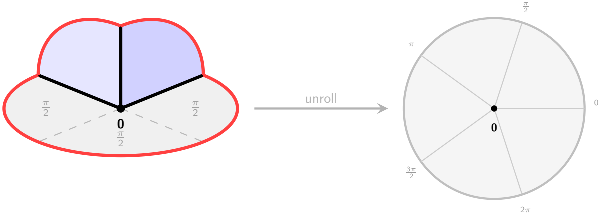
2D Stratified Spaces: Planes Glued Together



A finite number of planes (or wedges of planes) glued together along shared line segments form a **2D stratified metric space**.

Kales

A **kale** is composed of several half-planes glued together at a single point.

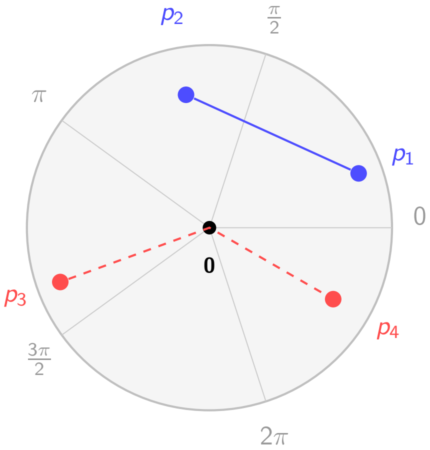


folded: hallway corner

unrolled: $\alpha = \frac{5\pi}{2}$

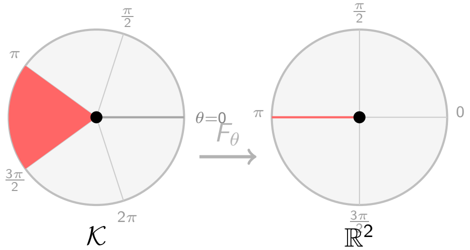
Here α is the **total angle sum** around $\mathbf{0}$: the ordinary plane has $\alpha = 2\pi$, while this kale is folded from $\alpha = \frac{5\pi}{2}$ — the extra angle is what makes $\mathbf{0}$ negatively curved.

The Distance Function on a Kale



$$\alpha = \frac{5\pi}{2}$$

Unfolding the Kale



Choose a direction θ and unfold the kale into the plane.

Each sampled point lands somewhere in $\mathbb{R}^2$ after unfolding.

The key quantity is the ordinary average of those unfolded points:

$$m_\theta = \mathbb{E}[F_\theta(X)].$$

The 2D Problem Becomes 1D

Suppose points are drawn in a radially symmetric way on the kale.

- X: a random point on the kale
- $Y = d(\mathbf{0}, X)$: how far X is from the center
- $Z = e_1 \cdot F_\theta(X)$: where X lands in the chosen direction after unfolding
- b_N: the Fréchet mean of the sample

Direction does not matter. The hard 2D geometry collapses to one question:

How far from the center is the mean?

Once we know the cost of landing at distance r, we know the cost of landing at any point p with $d(\mathbf{0}, p) = r$:

$$I_X(p) = I_Z\big(d(\mathbf{0}, p)\big).$$

Theorem: Features of a Kale

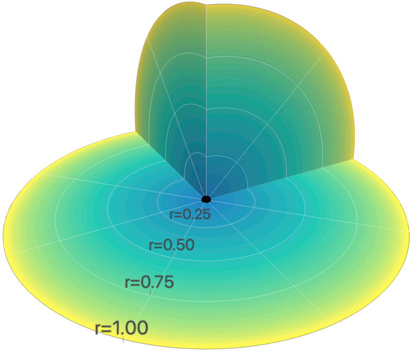
Suppose points are drawn in a radially symmetric way on the kale.

Theorem (Informal)

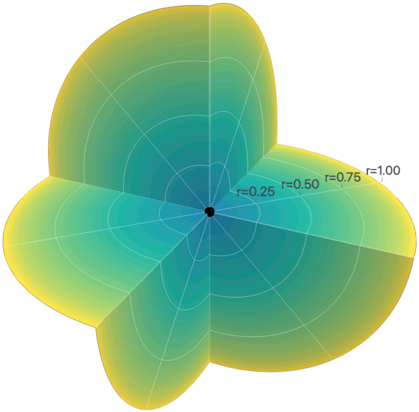
The rate function I_X of the Fréchet mean has the following features:

- *there is a* **jump** *at the center* $\mathbf{0}$*;*
- *adding more total angle makes deviations more costly;*
- *moving farther from the center becomes increasingly unlikely.*

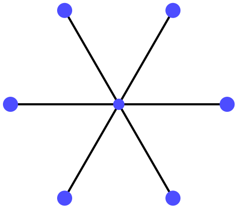
Example 1



Example 2



Future Directions



- Higher-dimensional stratified spaces
- Non-uniform distributions on vertices
- Continuous distributions on edges